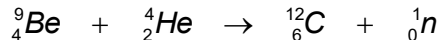


- 30** Consider the following nuclear reaction carried out by Chadwick, which led to the discovery of neutron.



Given that the binding energy per nucleon of ${}^{12}_6\text{C}$ is 7.67556 MeV, the binding energy per nucleon of ${}^9_4\text{Be}$ is 6.46276 MeV and the binding energy per nucleon of ${}^4_2\text{He}$ is 6.81854 MeV.

What is the energy change for this reaction?

- A** 5.60 MeV of energy is released.
- B** 5.60 MeV of energy is absorbed.
- C** 6.67 MeV of energy is released.
- D** 6.67 MeV of energy is absorbed.